

How close arms submitted and revised come to the 2011 targets

The model/target ratio of every calibration target at $t = 100$, for both arms and both regions; how the 28 September search for revised moved its instruments and ratios; whether revised's consumption is half of submitted's (it is not); and why the EU central bank holds no RoW bills while advances are negative.

LEEDS_MODEL — JUST2CE

2026-09-29

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Written: 2026-09-29 · **Repo at:** `main @ 8e07250`, plus the uncommitted `R/plot_calibration_ratios.R` and its outputs. **Reads:** the cached 100-period baselines `output/scenarios/arms/{submitted,revised,adjusted}` (cut 2026-09-23, 2026-09-29 07:35 and 2026-09-29 06:52), through the committed tables `output/calibration/calibration_ratio_cells_20260929.csv`, `consumption_levels_20260929.csv` and `advances_revised_20260929.csv`, which Rscript `R/plot_calibration_ratios.R` writes.

Which revised this describes. The calibrated baseline of 28 September, cut 2026-09-29 07:35, before the same day's restart fix (the money wage is now carried across the spin-up) and change R10. That run is kept as `output/scenarios/arms/revised/runs/baseline_pre_wagecar`. The cached **revised** baseline is now the fixed but **uncalibrated** one; see `qmd/reports/2026-09-29-equation-audit.qmd` §2 for how its ratios moved.

Summary

- **The targets.** `data/target_values.xlsx` holds twelve 2011 magnitudes per region (EU = Z1, RoW = Z2). Every ratio below is the model value at $t = 100$ divided by its target, in the model's unit of 10 bn EUR.
 - The three nominal concepts are value added, the deficit and the debt stock. They are divided first by the region's consumption deflator `pa`, indexed so that $t = 2 = 1$. This is `calib_fitness(deflate = TRUE)`, the convention every calibration search has used.
- **revised fits better overall.** The objective the searches minimised (`scale = "gdp"`, mean squared miss in units of GDP) is 0.00082 for **revised** against 0.01937 for **submitted**.

- **revised** puts value added, the debt stock, debt/VA and the deficit on target in both regions.
- **submitted** overshoots value added by 14–16% and the debt stock by 24% (EU) and 60% (RoW).
- **revised fits worse on three things:**
 - RoW real investment, 0.75 of target;
 - trade in both directions, 0.87–0.89 of target;
 - RoW gross output, 0.95.
- **Consumption is not half.** At $t = 100$ **revised**'s real consumption is 3.1% **above** **submitted**'s in the EU and 3.1% **below** it in RoW, and both are within 5.5% of target.
 - The impression came from the arm report `output/html/reports/arms/revised.html`, which plots each series indexed to its own value at $t = 2$.
 - The fixed-price arms start at $t = 2$ from a seed at about 43% of target and grow 2.4-fold. **revised** starts near its calibrated level and settles 9% lower.
 - Indexed, therefore, **revised** looks like about 0.38 of the other arm. That is a picture of two starting points, not of two consumption levels.
- **$b_cb_s_Z2 = 0$ is a design choice; negative advances are a sign convention.**
 - Change R6 freezes the EU central bank's RoW-bill stock at its $t = 1$ value, which is 0.
 - Change R7 defines a negative A_d as the banks' reserve deposit at the central bank. Both balance sheets and the gold-flow identity read the sign consistently.
- **Not a bug, but a defect: the banks make persistent losses.** The EU banks hold 959.3 of their 1,626.8 of deposits (59%) as reserves that earn nothing. RoW banks hold 49% that way.
 - Both pay a deposit rate above the bill rate.
 - Bank profit f_b is therefore negative in every period: -17.1 (EU) and -47.3 (RoW) at $t = 100$.

Figure A — the ratios at $t = 100$

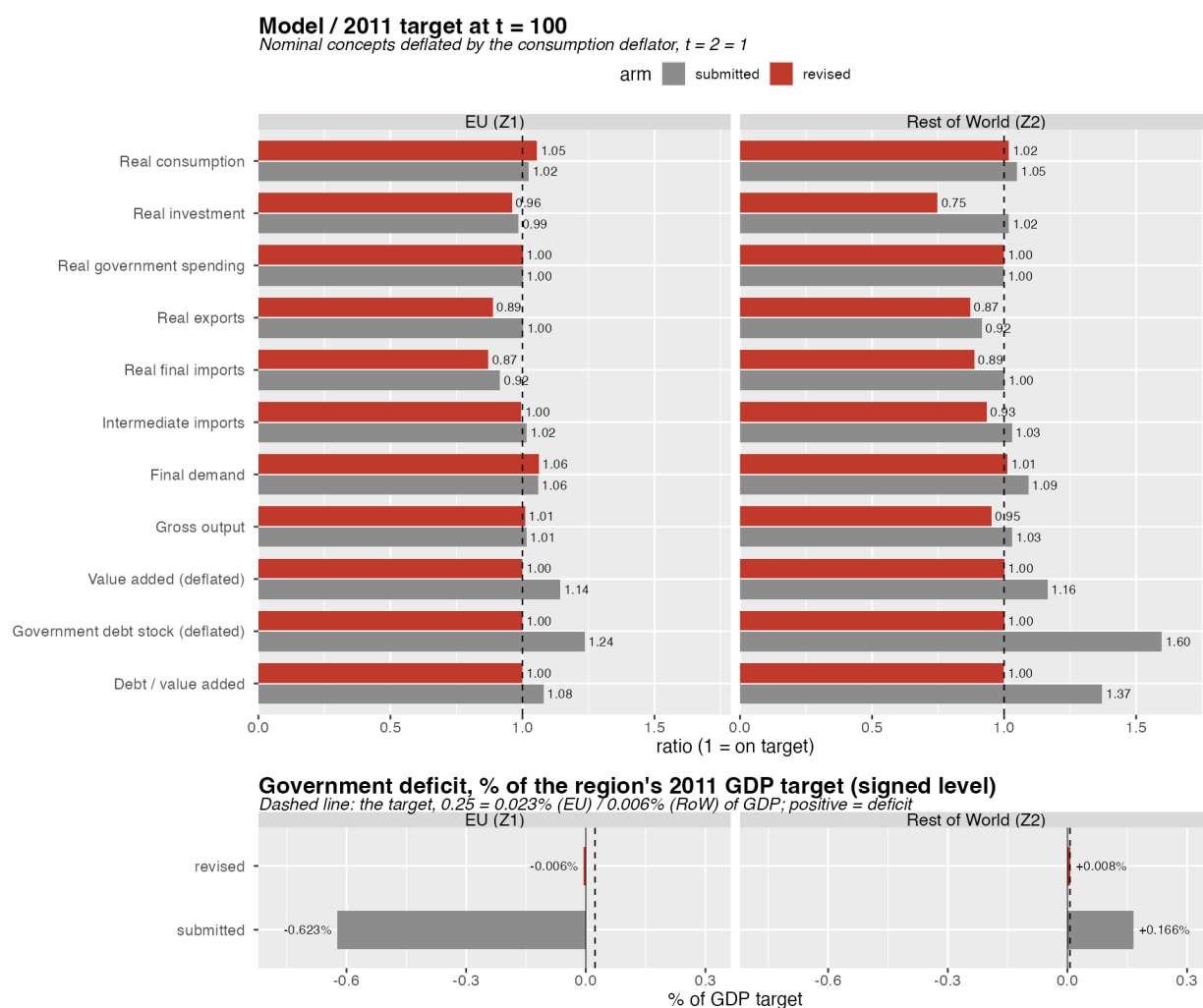


Figure 1: Model / 2011 target at $t = 100$. Bars: grey **submitted**, red **revised**. The deficit is drawn below it as a signed level in % of the region's GDP target, because its target (0.25, i.e. 2.5 bn EUR) is 0.023% of EU and 0.006% of RoW GDP, and a ratio to it (-27 for **submitted**) is not interpretable.

Table 1: Every target cell at $t = 100$. Units: 10 bn EUR, except debt/value added (a fraction). The deficit's ratio is shown for completeness only; read it in the figure as a share of GDP.

region	target concept	2011 target	submitted	ratio	revised	ratio
EU	Real consumption	626.32	640.97	1.023	660.63	1.055
EU	Real investment	196.62	193.67	0.985	188.82	0.960
EU	Real government spending	239.46	240.00	1.002	240.00	1.002
EU	Real exports	102.78	103.05	1.003	91.18	0.887
EU	Real final imports	66.36	60.76	0.916	57.78	0.871
EU	Intermediate imports	123.15	125.17	1.016	122.56	0.995
EU	Final demand	1098.81	1163.95	1.059	1167.24	1.062
EU	Gross output	2153.60	2185.73	1.015	2174.47	1.010
EU	Value added (deflated)	1081.51	1237.03	1.144	1082.06	1.001
EU	Government deficit (deflated)	0.25	-6.74	-26.972	-0.06	-0.239

region	target concept	2011 target	submitted	ratio	revised	ratio
EU	Debt stock (deflated)	876.03	1082.62	1.236	876.48	1.001
EU	Debt / value added	0.81	0.88	1.080	0.81	1.000
RoW	Real consumption	2370.71	2488.61	1.050	2412.12	1.017
RoW	Real investment	908.67	925.27	1.018	679.20	0.747
RoW	Real government spending	675.33	675.00	1.000	675.00	1.000
RoW	Real exports	66.36	60.76	0.916	57.78	0.871
RoW	Real final imports	102.78	103.05	1.003	91.18	0.887
RoW	Intermediate imports	105.85	108.95	1.029	98.84	0.934
RoW	Final demand	3918.29	4279.57	1.092	3968.53	1.013
RoW	Gross output	8424.52	8688.28	1.031	8022.84	0.952
RoW	Value added (deflated)	3935.58	4582.34	1.164	3941.14	1.001
RoW	Government deficit (deflated)	0.25	6.54	26.155	0.32	1.292
RoW	Debt stock (deflated)	3345.25	5343.53	1.597	3349.97	1.001
RoW	Debt / value added	0.85	1.17	1.372	0.85	1.000

Reading the table.

- **Exports and imports mirror each other by construction.** In a two-region model EU real exports are RoW real final imports, which is why the pairs 0.887 / 0.887 and 0.871 / 0.871 repeat across the two regions.
- **The trade shortfall.** In **revised**, trade is 11–13% below its 2011 level in both directions. Two causes are candidates, and neither is separated yet:
 - the Armington gap (hand-off item 2, defect 2);
 - the restart price jump (item 2, defect 3), which leaves Armington reference prices away from the stationary prices.
- **RoW investment at 0.75.** This is a property of **revised**'s capital block (R4, R9: investment is replacement demand $K \cdot x$ plus accumulation). The 09-28 calibration log recorded it as unfixed. The data's RoW net investment is 203 per period, so the model misses the gross level, not the net one.

Figure B — the search that calibrated revised

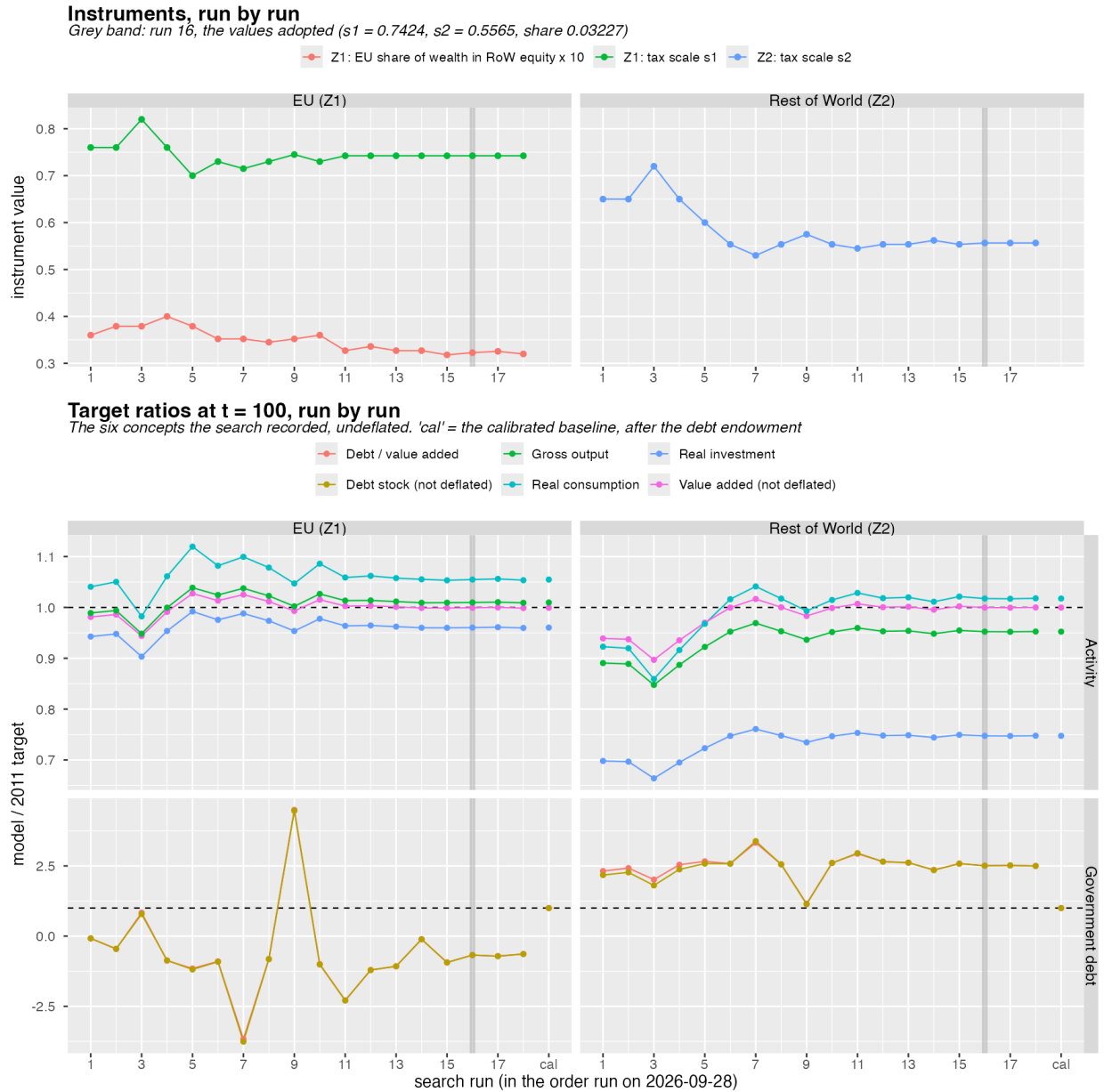


Figure 2: The 28 September joint search (output/calibration/revised_joint_search_20260928.csv), 18 runs in the order they were run, faceted by region. Top: the instruments. Bottom: the ratio to target of the six concepts the search recorded, split into activity and government debt. 'cal' is the calibrated baseline, cut after the search.

What the figure shows, stage by stage. revised was calibrated in three stages, and Figure B covers the middle one.

- Route A: the current account.** R/calibrate_revised_cab.R set the EU share of household wealth in RoW equity ($Z1_lambdaE0_2$, 0.02 in the workbook, not estimated) so that the EU current account is zero. This is what makes the baseline stationary.

- Output: output/calibration/revised_cab_search_2026092{8,9}.csv, 8 + 4 runs. They are not plotted, because they are one-dimensional.
- Result: 0.03714.

2. The joint search (Figure B), 18 runs.

- **Instruments:**

- a scale on the EU and RoW tax rates $_w$ and $_c$, which sets the level of activity (with a balanced budget, $VA = (G + \text{public investment} + \text{interest}) / \text{net tax rate}$);
- the same equity share, re-tuned jointly with the scale.
- **Adopted:** run 16, with scales 0.7424 (EU) and 0.5565 (RoW) and share 0.03227. From the 29 September commit 442f5f2 these are the workbook’s base values, so a scale of 1 is now the calibrated state.
- **Debt was not calibrated in this stage.** The debt endowment was 0 throughout, so debt/VA wandered: at run 16 it was -0.55 in the EU and 2.13 in RoW, ratios -0.67 and 2.5 .

3. The debt endowment. R/calibrate_revised_debt.R issued bills to each region’s own central bank at $t = 1$ of the spin-up: 1466.4 (EU) and -5045.7 (RoW).

- Why the instrument is an initial condition: under R6 debt and gold share a unit root once the current account is zero, so no flow parameter pins the debt level.
- Measured on 09-28: the endowment moves debt one for one and leaves value added, consumption and gross output identical to four digits.
- Figure B confirms it: the ‘cal’ points for the four activity concepts sit exactly on run 16, and debt/VA and the debt stock jump to 1.

What the search did not record. Real exports, final imports, intermediate imports, final demand, government spending and the deficit are not in the search file, so their evolution cannot be plotted without re-running the 18 configurations. The recorded levels are also undeflated: pa is not in the file. The final ratios of all twelve concepts are in Figure A.

Is revised’s consumption half of submitted’s?

No. The variable is Z_c , “Total Real Consumption”, in 10 bn EUR at base prices (data/Variable_Definitions.csv). Nominal consumption is $pa \times c$.

Table 2: Real consumption, its own index from $t = 2$, the consumption deflator and nominal consumption. **adjudicated** is the reference arm the per-arm reports plot **revised** against.

arm	region	c at t = 2	c at t = 100	c(100) / c(2)	pa at t = 100	nominal pa · c at t = 100	2011 target C
adjudicated	EU	270.9	650.5	2.401	1.070	696.2	626.3
adjudicated	RoW	971.7	2498.4	2.571	1.067	2666.1	2370.7
submitted	EU	270.4	641.0	2.370	1.070	686.0	626.3
submitted	RoW	971.1	2488.6	2.563	1.067	2655.6	2370.7
revised	EU	725.7	660.6	0.910	1.089	719.3	626.3
revised	RoW	2548.2	2412.1	0.947	1.188	2865.1	2370.7

- **In volume at $t = 100$:**

- EU: revised 660.6 against submitted 641.0, +3.1%;
- RoW: 2,412.1 against 2,488.6, -3.1%.

- **In money:** revised is higher in both regions, because its deflator is higher.

- pa is 1.089 (EU) and 1.188 (RoW) against 1.070 and 1.067.
- The reason is that R1 measures prices of production in money units.

- **So a price-level difference does not hide a volume difference here.** Both arms deflate c the same way, and their volumes differ by 3%.
- **Where “half” came from.** The arm report (`qmd/reports/arms/arm-baseline.qmd`, rendered as `output/html/reports/arms/revised.html`) plots `revised` against `adjudicated`, with every series divided by its own value at $t = 2$.
 - `adjudicated`’s EU consumption runs from 270.9 to 650.5, an index of 2.40, because the fixed-price arms start from the old seed at 43% of target.
 - `revised`’s starts at 725.7 and ends at 660.6, an index of 0.91.
 - On that panel `revised` therefore sits at $0.91 / 2.40 = 0.38$ of `adjudicated`.
- **Why revised starts high.** Its $t = 2$ value is itself the restart overshoot: c is 701.4 at $t = 1$, 725.7 at $t = 2$ and 659.3 at $t = 3$. That is the same period as the 6–18% price jump at the spin-up restart (hand-off item 2, defect 3). So the index base is the worst possible period for this arm.
- **Recommendation.** The arm reports should plot levels, or index to the target, whenever the two arms start from different initial states. Not changed here; a one-line option in `arm-baseline.qmd`.

Why is `b_cb_s_Z2` 0 while advances are negative?

The two facts are unrelated.

`Z1_b_cb_s_Z2` is the EU central bank’s stock of RoW bills. It is 0 in every period because R6 (G&L OPENFIX) freezes it at its previous value (`model/code/MVP_model_2026.R:1947`, `sim['Z1_b_cb_s_Z2', i] <- sim['Z1_b_cb_s_Z2', i - 1]`), and its $t = 1$ value is 0.

- The freeze is deliberate. It cuts the RoW treasury $\rightarrow$ EU central bank $\rightarrow$ EU treasury interest circuit that compounded at $1 + r_b$ in the older arms (engine comment at lines 1935–1946).
- The workbook’s `b_cb_d_Z2` is among the 27 labels `revised` never reads (`output/calibration/workbook_us`).
- Under OPENFIX the balance of payments settles in gold or, not in this stock. That is the closure’s defining choice.

`a_d` is the banks’ demand for central-bank advances. Under R7 (`MVP_model_2026.R:683`) banks demand bills as a fixed share of deposits, `b_b = bank_bill_ratio * M_s` (0.1 in both regions), and advances clear the bank balance sheet:

$$A_d = L_s + B_b - M_s.$$

When deposits exceed loans plus bills, `A_d` is negative. The engine comment at line 187 states the convention: “negative = reserve deposits at the CB”.

The sign is used consistently everywhere it appears. Checked:

- **Bank balance sheet.** Assets $L + B_b$, liabilities $M + A$. A negative A is an asset of the banks.
- **Central-bank balance sheet.** Supply `a_s = a_d` (line 857). The gold-flow identity (line 1957) subtracts Δa_s from ΔH . A fall in A , i.e. a rise in reserves, is therefore a CB liability that adds to its reserve assets, which is the right direction.
- **World gold conservation.** `run.model()`’s consistency check measures it, and it closes (the 09-28 calibration recorded a consistency error of $4.3e-9$).

What is a defect: those reserves earn nothing, and the banks lose money.

Table 3: Bank balance sheet and rates in the **revised** baseline (10 bn EUR; rates per period).

region	variable	t1	t2	t100
EU	ms	1549.3657	1576.4643	1626.7551
EU	ls	456.9478	503.8555	504.8164
EU	b_b	154.9366	157.6464	162.6755
EU	a_d	-937.4813	-914.9623	-959.2632
EU	f_b	-17.0868	-17.0721	-17.1350
EU	rm	0.0226	0.0226	0.0226
EU	rl	0.0323	0.0323	0.0323
EU	rb	0.0216	0.0216	0.0216
RoW	ms	7343.2002	7562.5213	8644.1715
RoW	ls	3314.5633	3660.8591	3580.0033
RoW	b_b	734.3200	756.2521	864.4172
RoW	a_d	-3294.3168	-3145.4101	-4199.7511
RoW	f_b	-31.4998	-31.7458	-47.3380
RoW	rm	0.0202	0.0202	0.0202
RoW	rl	0.0291	0.0291	0.0291
RoW	rb	0.0217	0.0217	0.0217
EU	b_cb_s_Z2	0.0000	0.0000	0.0000

- **The balance sheet.** At $t = 100$ the EU banks hold 1,626.8 of deposits against 504.8 of loans and 162.7 of bills, so 959.3 sits at the central bank.
- **Why they lose money.** Bank profit (line 700) is $f_b = r_l L_f + r_b B_b - r_m M_s + r_h L_h$. It has **no term in A**, so the reserves earn nothing.
 - The deposit rate **rm** is 0.0226, above the bill rate **rb** 0.0216.
 - **f_b** is therefore negative in every period: -17.1 (EU) and -47.3 (RoW).
- **This is consistent accounting of an implausible configuration.** As recalled, not yet checked against the book: in G&L’s ch. 10 bank model reserves are a required fraction of deposits and bills are the residual asset, so banks never sit on 59% of their deposits as idle reserves.
- **Two candidate fixes, for a decision:**
 1. calibrate **bank_bill_ratio** so that **A_d** = 0, making bills rather than reserves the residual asset. That is roughly 0.69 (EU) and 0.59 (RoW) at the $t = 100$ balance sheets, since $(M - L)/M = 0.69 / 0.59$;
 2. pay the policy rate on reserves, adding **ra** $\times$ **A** to both **f_b** and **f_cb**.
- **Neither is made here.** Either changes **revised**’s calibrated state and needs a re-cut.

Found in passing

- **CLAUDE.md is out of date on one fact.** It says the gold endowment is “1,500 each, inert”. The workbook holds **Z1_gold_endowment** = 1,500 and **Z2_gold_endowment** = **2,500**, raised on 09-28 because RoW gold ended at -586 (session log 2026-09-29 §6).
- **Search file headers.** The joint-search file’s instrument columns are **d1**, **d2**, **lambda**, **s1** and **s2**. **d1** and **d2** are the debt endowments, 0 in all 18 runs, and **s1/s2** are the tax scales. The header does not say this; the script’s comment (**R/calibrate_revised_debt.R:15**) does.

Reproduce

```
Rscript R/plot_calibration_ratios.R      # figures + three CSVs, from the cached baselines (~10  
quarto render qmd/reports/2026-09-29-calibration-ratios.qmd --to html --output-dir ../../outp
```